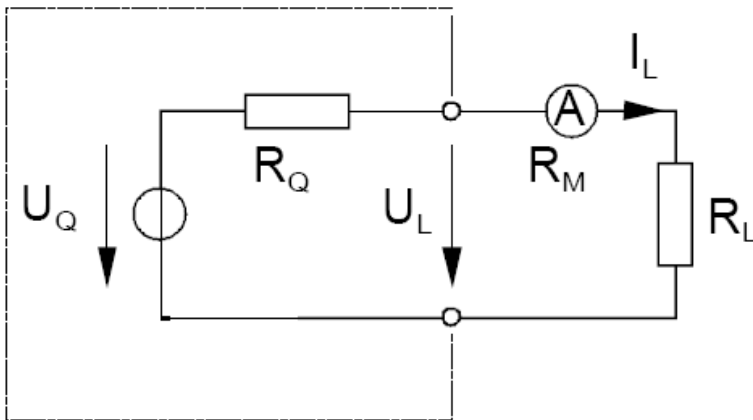


5. DC Measurements

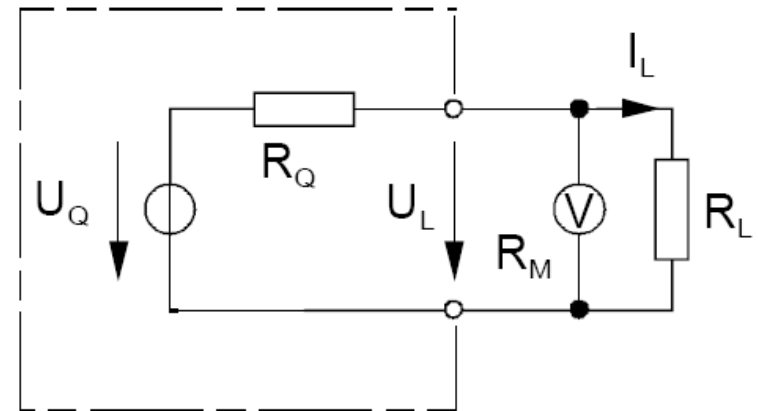
Measurement of direct current and direct voltage

- the measuring device in the circuit changes the quantity to be measured (systematic error)
- the measuring device is part of the system and thus changes the system!



current measurement

Measured value I_L is systematically too small



voltage measurement

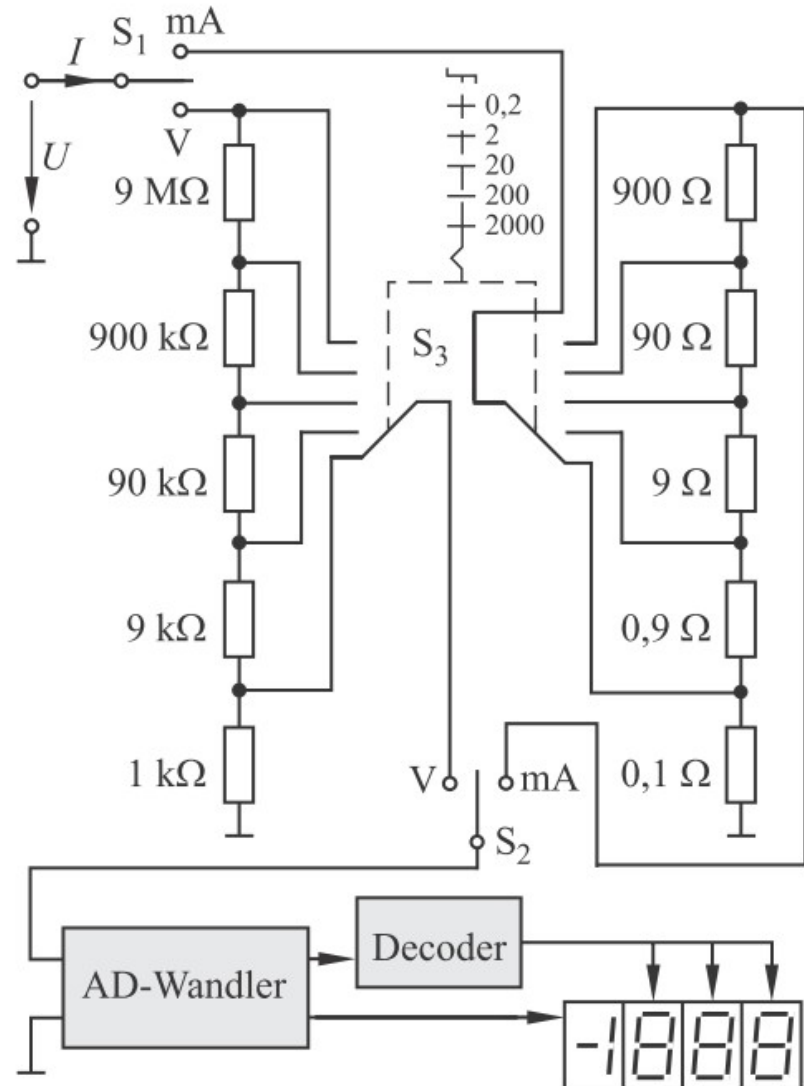
Measured value U_L is systematically too small

- current measurement: $R_M \ll R_Q + R_L$
- voltage measurement: $R_M \gg R_L$

5. DC Measurements

common circuit for digital multimeters

- current measurement
input resistance depends on range
- voltage measurement
input resistance independent of range



5. DC Measurements

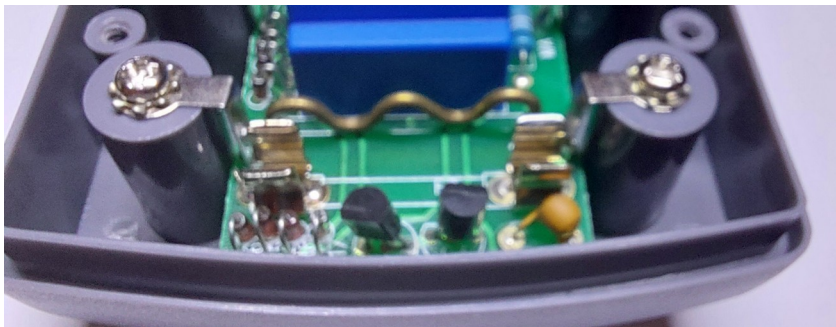
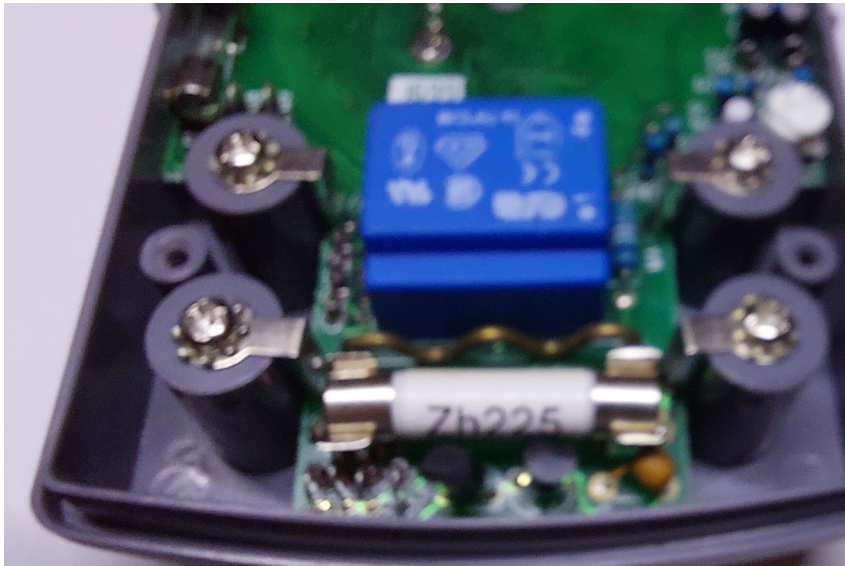


COM

mA

20A

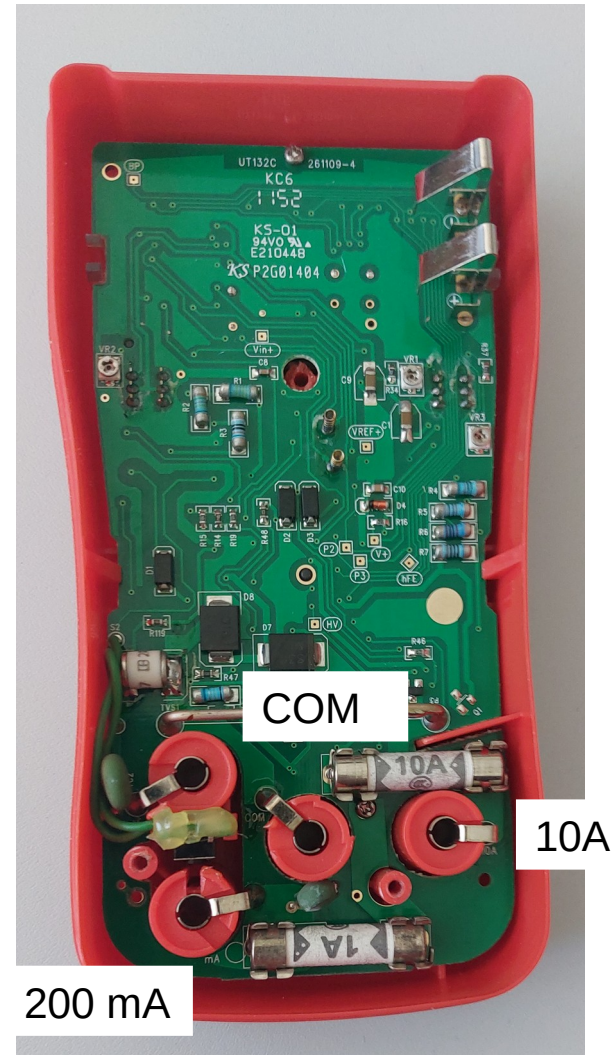
5. DC Measurements



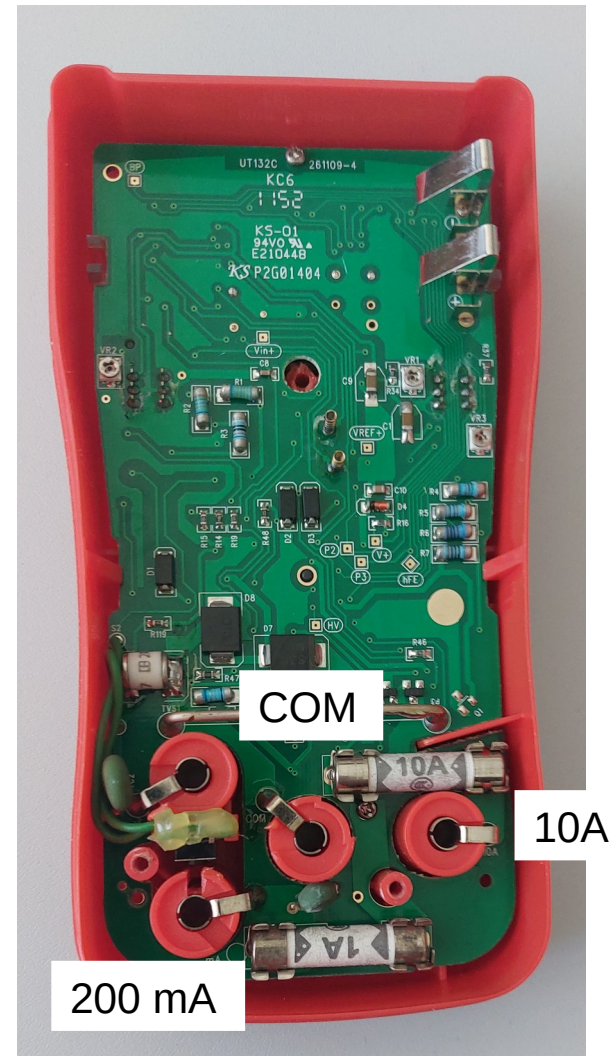
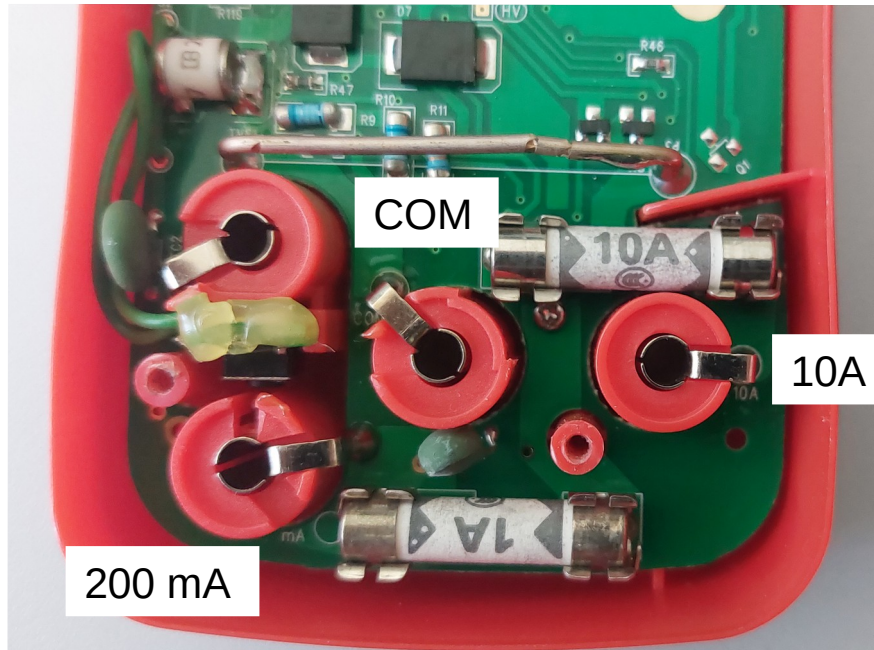
mA

20A

5. DC Measurements



5. DC Measurements

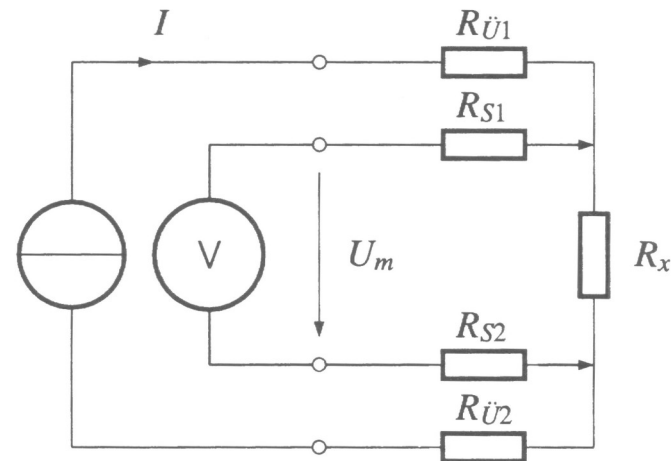
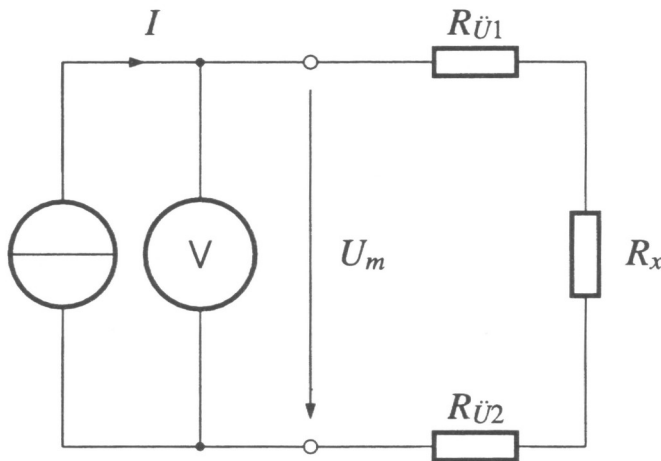


5. DC Measurements

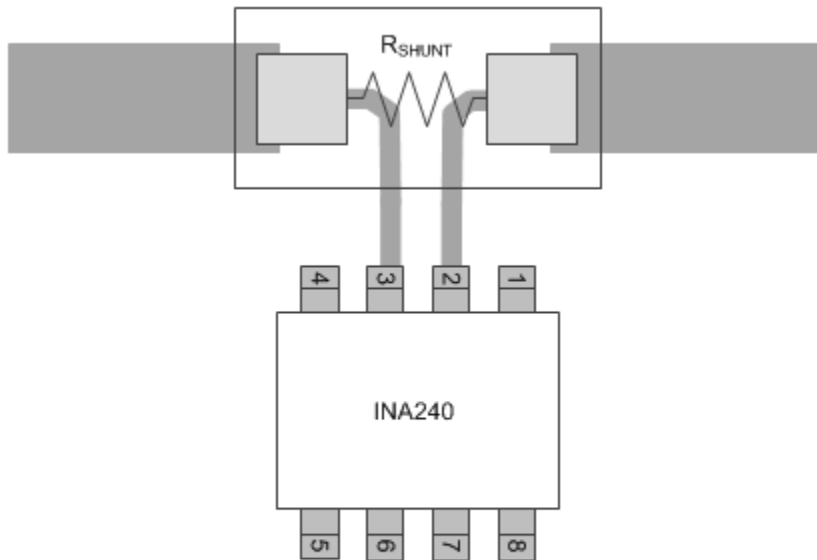
Resistance measurement

- resistor is operated e. g. at a constant current source with I_0
- 2-wire measurement (standard)
- **4-wire measurement** with separate leads for supply and measurement for very small resistances, to eliminate the effect of the measuring leads

sense-leads (“Kelvin clamps”) are used to measure the voltage drop across the resistor independently of the voltage drop along the supply leads

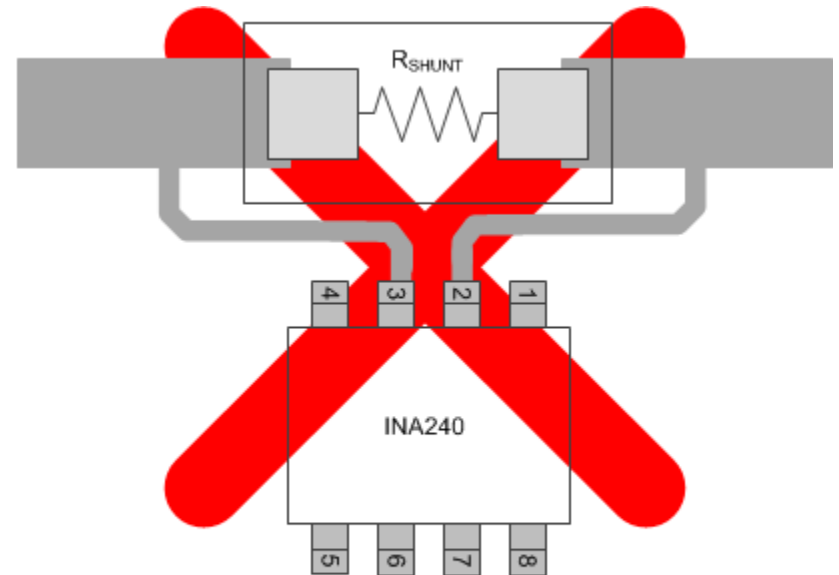


5. DC Measurements



DO

Kelvin Connection from Shunt Resistor



DON'T

Non-Kelvin Connection from Shunt Resistor

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